Lecture 3: Integration

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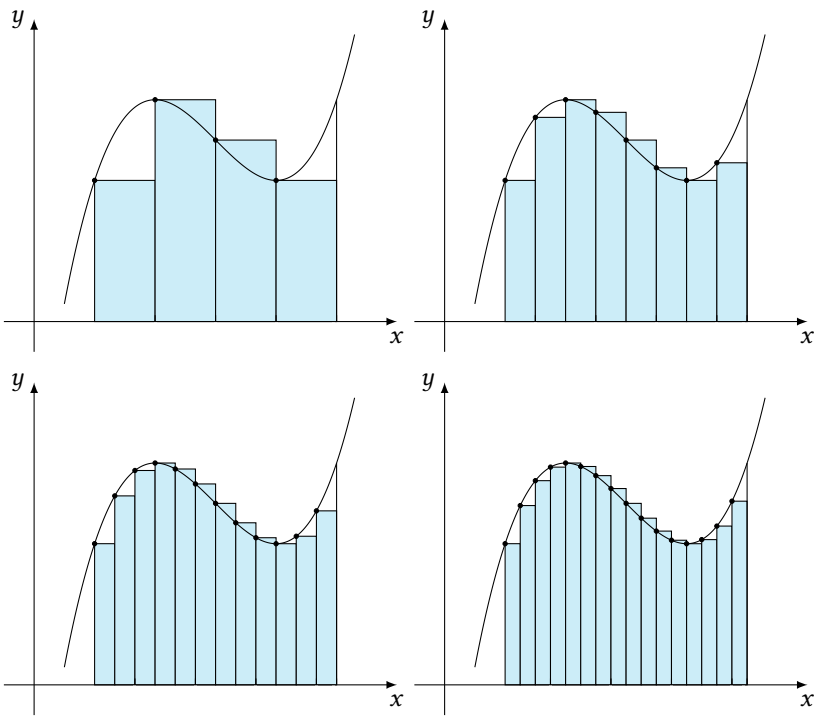
Recall the area problem: given any polygon, we are able to find its area because we can cut it into triangles, whose individual areas can be measured. Given a region enclosed by curves, however, we do not have an explicit formula to approximate its area.

Question. *How do we find the area of the region under a curve $y = f(x)$, over an interval $a \leq x \leq b$?*

We can use areas of known polygons—rectangles, for instance—to approximate the area under a curve.



We divide the interval $[a, b]$ into n subintervals of size Δx, say

$$[x_0, x_1], \ [x_1, x_2], \ \ldots, \ [x_{n-1}, x_n], \quad \text{with} \quad \Delta x = \frac{b-a}{n}.$$

Then, we can estimate the height of the rectangle at the interval $[x_{i-1}, x_i]$ with $f(x_i)$. We can thus approximate the *area* A of the region under the graph as a limit of a **Riemann sum**:

$$A = \lim_{n\to\infty} \left[f(x_1)\,\Delta x + f(x_2)\,\Delta x + \cdots + f(x_n)\,\Delta x \right].$$

This limit is the definition of the **definite integral** of $f(x)$ from a to b:

$$\int_a^b f(x)\ dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x.$$

The function $f(x)$ is called the **integrand**, and the values a and b are the **limits of integration**. We thus have a geometric interpretation of the definite integral: it is the *signed area* of the region between the graph of $y = f(x)$ and the x-axis.

Properties of the Definite Integral

We now list some properties of the definite integral.

- Swapping the limits of integration changes the sign of the integral:

$$\int_b^a f(x)\ dx = -\int_a^b f(x)\ dx.$$

- There is no area under a single point:

$$\int_a^a f(x)\ dx = 0.$$

- The region under a horizontal line is simply a rectangle:

$$\int_a^b c\, dx = c(b-a).$$

- The integral is a linear operator:

$$\int_a^b [f(x)+g(x)]\ dx = \int_a^b f(x)\ dx + \int_a^b g(x)\ dx,$$

$$\int_a^b cf(x)\ dx = c\int_a^b f(x)\ dx.$$

- We can combine integrals of the same function over adjacent intervals:

$$\int_a^c f(x)\ dx + \int_c^b f(x)\ dx = \int_a^b f(x)\ dx.$$

The Fundamental Theorem of Calculus

We can consider the integral as a function by evaluating

$$g(x) = \int_a^x f(t)\ dt.$$

We can think of this as a function that gives us the area between the graph of $y = f(x)$ and the x-axis from a to x, where x can vary—the "area so far", so to speak.

We now have one of the most important results in mathematics, uniting the two distinct branches of differential calculus (which investigates tangents) and integral calculus (which computes areas).

Theorem (Fundamental Theorem of Calculus, I)**.** *Let $f(x)$ be continuous on an interval $[a,b]$. Then, the function*

$$g(x) = \int_a^x f(t)\ dt, \quad a \le x \le b$$

is continuous on $[a,b]$ and differentiable on (a,b), with

$$g'(x) = f(x).$$

In other words, differentiation and integration are inverses of one another:

$$\frac{d}{dx}\int_a^x f(t)\ dt = f(x).$$

Theorem (Fundamental Theorem of Calculus, II)**.** *Let f be continuous on $[a, b]$, and let $F(x)$ be any* ***antiderivative*** *of $f(x)$: that is, a function such that $F'(x) = f(x)$. Then,*

$$\int_a^b f(x)\ dx = F(x)|_{x=a}^{x=b} = F(b) - F(a).$$

Example. Consider

$$\int_1^2 e^x\, dx.$$

By the Fundamental Theorem of Calculus, it suffices to determine an antiderivative of e^x: this is fairly straightforward, since

$$\frac{d}{dx} e^x = e^x.$$

Thus,

$$\int_1^2 e^x\, dx = e^x|_{x=1}^{x=2} = e^2 - e.$$

From these results, the problem of evaluating definite integrals is reduced to finding the antiderivatives of the integrand. To denote the antiderivative of $f(x)$, we thus introduce the notation of an **indefinite integral**

$$\int f(x)\ dx.$$

It is thus clear that

$$F'(x) = f(x) \quad \Longleftrightarrow \quad \int f(x)\ dx = F(x) + c.$$

The rules of differentiation thus extend to integration:

- $\displaystyle\int k\, dx = kx + c,$
- $\displaystyle\int x^n\, dx = \frac{x^{n+1}}{n+1} + c$ for $n \neq -1$,
- $\displaystyle\int \sin x\, dx = -\cos x + c,$
- $\displaystyle\int \cos x\, dx = \sin x + c,$
- $\displaystyle\int \sec^2 x\, dx = \tan x + c$, and so on.

Furthermore, note that the indefinite integral is a linear operator:

$$\int (af(x) + bg(x))\ dx = a\int f(x)\ dx + b\int g(x)\ dx.$$

Example. Let us compute

$$\int 10x^4 - 2\sec^2 x\, dx.$$

Observe that

$$\begin{aligned}\int 10x^4 - 2\sec^2 x\, dx &= 10\int x^4\, x - 2\int \sec^2\ dx \\ &= 10 \cdot \frac{x^5}{5} - 2\tan x + c \\ &= 2x^5 - 2\tan x + c.\end{aligned}$$

Integration by Substitution

Oftentimes, integrals are difficult to compute directly, and a substitution may be required to transform it into a integral whose antiderivatives we are able to find

Example. Let us consider

$$\int 2x\sqrt{1-x^2}\,dx.$$

None of the standard formulae allow us to solve this integral directly, so we consider a substitution: let

$$u = 1 - x^2 \quad \Longrightarrow \quad \frac{du}{dx} = -2x \quad \Longrightarrow \quad du = -2x\,dx.$$

Then, we may rewrite the integral in terms of u:

$$\int 2x\sqrt{1-x^2}\,dx = \int \left(-\sqrt{u}\right)\,du = -\int u^{1/2}\,du = -\frac{u^{3/2}}{3/2} + c.$$

Substituting back $u = 1 - x^2$, we have

$$\int 2x\sqrt{1-x^2}\,dx = -\frac{2}{3}\left(1-x^2\right)^{3/2} + c.$$

The main idea in this example is to observe that a substitution can be made if a function $g(x)$ and its derivative $g'(x)$ both show up in the integrand: letting $u = g(x)$, we have

$$\int f(g(x))\,g'(x)\;dx = \int f(u)\;du.$$

Note that this result arises from the *Chain Rule* of differentiation.

Example. We compute

$$\int x^3\cos\left(x^4+2\right)\;dx.$$

Letting $u = x^4 + 2$, with $du = 4x^3\,dx$, we may rewrite the integral and evaluate it:

$$\int \frac{1}{4}\cos u\,du = \frac{1}{4}\sin u + c = \frac{1}{4}\sin\left(x^4+2\right) + c.$$

Example. Consider the integral of the tangent function

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx.$$

Let $u = \cos x$, with $du = -\sin x\,dx$. Then,

$$\int \frac{\sin x}{\cos x}\,dx = \int -\frac{1}{u}\,du = -\ln|u| + c = -\ln|\cos x| + c.$$

We may encounter other types of substitutions as well: remember that the goal is to transform the integrand into a form whose integral can be evaluated.

Example. Consider the integral

$$\int \frac{1}{1+\sqrt{x}}\,dx.$$

The term $\sqrt{x}$ in the denominator is somewhat difficult to deal with: thus, we let $u = \sqrt{x}$, or $x = u^2$, such that $dx = 2u\,du$. Then, we may rewrite the integral as

$$\int \frac{1}{1+u} \cdot 2u\,du = 2\int \frac{u}{1+u}\,du.$$

We rewrite the integrand by 'adding zero' to the numerator:

$$\begin{aligned} 2\int \frac{u}{1+u}\,du &= 2\int \frac{u+1-1}{1+u}\,du \\ &= 2\int 1 - \frac{1}{1+u}\,du. \end{aligned}$$

We can thus evaluate this integral term by term:

$$2\int 1 - \frac{1}{1+u}\,du = 2\left(u - \ln|1+u|\right) + c.$$

Since $u = \sqrt{x}$, we conclude that

$$\int \frac{1}{1+\sqrt{x}}\,dx = 2\left(\sqrt{x} - \ln\left(1+\sqrt{x}\right)\right) + c.$$

Example. Let us use the substitution $x = \sin t$* to evaluate the integral

$$\int \sqrt{1-x^2}\,dx.$$

* This is an example of a trigonometric substitution (see *Extra Notes on Integrals*).

Observe that, by the Pythagorean Theorem, we have

$$\cos^2 t + \sin^2 t = 1 \quad\Longrightarrow\quad \sqrt{1-x^2} = \sqrt{1-\sin^2 t} = \cos t.$$

Note, as well, that that $dx = \cos t\,dt$. The integral thus becomes

$$\int \sqrt{1-x^2}\,dx = \int \cos t \cos t\,dt = \int \cos^2 t\,dt.$$

By the half-angle identities of trigonometry, this is simply

$$\int \cos^2 t\,dt = \int \frac{1}{2}\left(1+\cos 2t\right)\,dt = \frac{1}{2}\left(t + \frac{1}{2}\sin 2t + \right) c.$$

Since $x = \sin t$, we have $t = \sin^{-1} x$. Furthermore, note that

$$\sin 2t = 2\sin t \cos t = 2x\sqrt{1-x^2}.$$

Hence, we conclude that

$$\int \sqrt{1-x^2}\,dx = \frac{1}{2}\sin^{-1} x + \frac{1}{2}x\sqrt{1-x^2} + c.$$

Integration by Parts

The techniques of integration often arise from what we know about differentiation: recall the *Product Rule*:

$$\frac{d}{dx}\left[f(x)\,g(x)\right] = f'(x)\,g(x) + f(x)\,g'(x)\,.$$

By the Fundamental Theorem of Calculus, we may write this with indefinite integrals:

$$f(x)\,g(x) = \int f'(x)\,g(x)\;dx + \int f(x)\,g'(x)\;dx.$$

We thus arrive at the **integration by parts** formula:

$$\int f(x)\,g'(x)\;dx = f(x)\,g(x) - \int f'(x)\,g(x)\;dx.$$

Taking $u = f(x)$ and $v = g(x)$, we may write this succinctly as

$$\int u\,dv = uv - \int v\,du.$$

Example. Let us use integration by parts to evaluate

$$\int x \sin x\,dx.$$

Our first job is to split the integrand into corresponding parts u and dv: let $u = x$ and $dv = \sin x\,dx$. Then,

$$u = x \implies du = dx, \qquad dv = \sin x\,dx \implies v = -\cos x.$$

Plugging these directly into the integration by parts formula, we have

$$\int x \sin x\,dx = x(-\cos x) - \int -\cos x\,dx = -x\cos x + \sin x + c.$$

Integration by parts is often useful when we wish to integrate difficult functions with known derivatives.

Example. Let us evaluate

$$\int \ln x\,dx.$$

We let the parts be

$$u = \ln x \implies du = \frac{1}{x}dx, \qquad dv = dx \implies v = x.$$

Then,

$$\int x \ln x\,dx = \ln x \cdot x - \int x\,\frac{1}{x}\,dx = x\ln x - x + c.$$

Extra Notes on Integration

Trigonometric Substitution. Suppose we wish to integrate functions of the form

$$\sqrt{a^2 - x^2}, \quad \sqrt{a^2 + x^2}, \quad \sqrt{x^2 - a^2}.$$

The *Pythagorean identities* give us clues as to substitutions that we may use to more evaluate these integrals. We recall the two key identities:

$$\sin^2 t + \cos^2 t = 1, \quad \tan^2 t + 1 = \sec^2 t.$$

Then:

- to evaluate integrals involving $\sqrt{a^2 - x^2}$, try

$$x = a \sin t, \quad -\pi/2 \leq t \leq \pi/2;$$

- to evaluate integrals involving $\sqrt{a^2 + x^2}$ or $a^2 + x^2$, try

$$x = a \tan t, \quad \pi/2 < t < \pi/2;$$

- to evaluate integrals involving $\sqrt{x^2 - a^2}$, try

$$x = a \sec t, \quad 0 \leq t < \pi/2 \text{ or } \pi \leq t < 3\pi/2.$$

Example. Consider the integral

$$\int \frac{\sqrt{9 - x^2}}{x^2}\, dx.$$

We perform a trigonometric substitution: let $x = 3 \sin t$, with $dx = 3 \cos t\, dt$, where $-\pi/2 \leq t \leq \pi/2$. Then,

$$\sqrt{9 - x^2} = \sqrt{9 - 9 \sin^2 t} = \sqrt{9 \cos^2 t} = 3 \left|\cos t\right|.$$

Since $-\pi/2 \leq t \leq \pi/2$, we have $\cos t \geq 0$ and so our integral becomes

$$\int \frac{\sqrt{9 - x^2}}{x^2}\, dx = \int \frac{3 \cos t}{9 \sin^2 t} \cdot 3 \cos t\, dt = \int \cot^2 t\, dt.$$

Since $1 + \cot^2 t = \csc^2 t$, we hav

$$\int \cot^2 t\, dt = \int \csc^2 t - 1\, dt = -\cot t - t + c.$$

Since $x = 3 \sin t$, we have $t = \sin^{-1}(x/3)$. Furthermore, note that

$$\cot t = \frac{\cos t}{\sin t} = \frac{\sqrt{9 - x^2}}{x}.$$

Hence, we conclude that

$$\int \frac{\sqrt{9 - x^2}}{x^2}\, dx = -\frac{\sqrt{9 - x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + c.$$